Bradley L. Davis

Electrical Hardware Engineer

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Professional Experience

Schweitzer Engineering Laboratories // Pullman, WA

Lead Hardware Engineer // January 2025 - Present

- Developed and maintained company-wide tools using Flask and HTMX that serve a holistic view of product lifecycle analysis. Aggregated incongruent datasets that saves time for >1000 daily users.
- Contributed to process improvements between R&D and in-house manufacturing divisions. Increased manufacturability and decreased time to market.
- Created hardware and software for testing 10 Gigabit Ethernet devices and PoE sources. Included PoE electronic load circuit, microcontroller-based analog voltage monitor, and full-stack web development. Streamlined setup and pre/post-testing for regulatory type testing.

Hardware Engineer // July 2022 - December 2024

- Led development of GNSS-disciplined clock enhancement to add new timing signals with nanosecond accuracy. Collaborated directly with the customer to understand requirements and prototyping circuits in KiCAD.
- Mentored & supervised engineering interns.
- Automated test equipment using Python and VISA to reproducibly analyze units for functional performance.

Associate Hardware Engineer // July 2021 - June 2022

- Qualified electronic components that meet or exceed electrical and regulatory equivalence.
- Utilized oscilloscopes, digital multimeters, and network analyzers to functionally test hardware.
- Evaluated PCB layout in ANSYS SIwave simulations and recommended alterations that improved signal integrity and EMC.

Hardware Engineering Intern // May 2018 - June 2021

- Aided development for an Intel-based rugged industrial computer for real-time automation.
- Performed UL, IEC, & IEEE regulatory type testing including surge, EFTB, EMC, thermal cycling, & dielectric strength.
- Developed a Gigabit Ethernet reliability and throughput testing software using C++.
- Fabricated a testing tool that identified manufacturing defects. Saving money and improves product safety.

Education

Washington State University

Fall 2017 - Fall 2021

B.S. Electrical Engineering

Magna Cum Laude

Math & Physics minors

Club Experience

Cougs in Space

Systems lead for cubesat design & development

Personal Projects

homelab // 2018 - Present

Learned system administration with self-hosted services.

Deployed with Ansible: NGINX, pfSense, chrony, TrueNAS, Pi-hole, Uptime Kuma, Docker, Proxmox, Grafana, Prometheus, Certbot, DDNS, & more.

nummus // 2022 - Present

Utilized Flask, sqlite, tailwindcss, Chart.js to create a personal financial information aggregator and planning tool.

Prototyped technology stacks to feedback into professional projects.